Shuqi Ke

kkksukie@gmail.com | 15707039996 | ksukie

Employment

Shenzhen Shita Robotics Technology Co., Ltd. *Shenzhen, China*
Algorithm Engineer Intern *Mar. 2026 - Aug. 2026*

- Built a real-robot visual-tactile data collection, recording, training, and inference pipeline for embodied intelligence and edge vision; integrated Piper leader-follower control, LTeach tactile cameras, and multimodal data management; adapted ACT and PI0 / PI0.5 training configurations and effective batch-size accounting.
- In parallel, maintained a visual-tactile optical-flow estimation SDK, optimized hardware solutions, and deployed algorithms; developed two-frame motion-vector prediction, multi-camera detection, and template tracking; implemented FP16 / PTQ INT8 quantization, calibration-set construction, and PC/RV1126B/RK3588 consistency validation with GStreamer/MIPI pipelines, concurrent processing, profiling, and on-site calibration.

Education

Jiujiang University *Jiujiang, China*
IoT Engineering *Sep. 2022 - Jun. 2026*

- **Core Coursework**: Data Structures, Computer Networks, Operating Systems, Computer Organization, Advanced Programming.
- **Language Certifications**: CET-4 and CET-6.

Research Experience

CVI Lab, Jiujiang University
Undergraduate Research Assistant · Computer Vision *Dec. 2022 - Sep. 2025*

- Conducted independent and collaborative research in object detection, medical image segmentation, and edge AI; handled data annotation and processing; investigated ResNet and DenseNet backbones and segmentation models such as SAM; designed experiments, implemented models, performed comparative, ablation, and error analyses; and contributed to wire-sequence model research, manuscript preparation, and experimental work for a first-author paper.
- Developed and maintained a PyQt5-based end-to-end biomedical imaging workflow for chromosome preprocessing, candidate-region detection, result review, abnormal-sample annotation, batch loading, manual correction, and structured result export; responsible for UI development and related front-end/back-end maintenance.

Research Interests

Tactile perception, multimodal robot learning, and algorithm optimization for contact-rich manipulation, spanning visual-tactile data collection, multimodal policy training, tactile representation and force-field modeling, sim-to-real transfer, and edge-device deployment; progressively exploring world models for embodied intelligence.

Publications

[1] TY-YOLO: An Attention-Based Multi-Scale Complex Scene Fire Smoke Detection Algorithm [Under Review]
First Author · Pattern Recognition Letters

- Led model selection, architecture optimization, loss-function tuning, and manuscript preparation; combined attention mechanisms with multi-scale feature fusion to address small-target smoke, occlusion, and false positives in complex backgrounds.

[2] FgFEU-Net: A Fine-Grained Feature Extraction U-Net Model for Breast Tumor Segmentation Based on Transfer Learning [Paper]
Breast Ultrasound Segmentation · Co-author *EDGE 2025 · Springer LNCS 16154 (2026)*

- Developed FgFEU-Net, a transfer-learning U-Net integrating a VGG16 encoder, same-level feature fusion, and Combo Loss to improve tumor-boundary delineation and mitigate foreground–background imbalance in breast ultrasound; achieved 99.03% accuracy, 96.83% precision, and 99.35% sensitivity on BUS2018.

[3] The Real-Time Intelligent Transportation Detection System Based on Edge-Cloud Collaborative Computing [Paper]
Edge–Cloud Intelligent Transportation · Co-author *EDGE 2025 · Springer LNCS 16154 (2026)*

- Deployed YOLOv8n on edge devices for vehicle recognition, traffic-flow analysis, congestion monitoring, wrong-way driving detection, and boundary-crossing alerts; integrated edge–cloud collaboration to offload centralized processing, achieving 55.06 FPS and an F1 score of 0.84 on edge hardware.

Project Portfolio

VTLA *LeRobot · PI0.5 · Piper*

VLA / Robot Learning

- Extended LeRobot 0.5.2 with Piper leader-follower control, LTeach tactile sensing, and LeRobot-compatible recording, training, and inference workflows while preserving the upstream CLI/API.
- Implemented dedicated tactile pathways for PI0.5 and ACT-based DreamTacVLA, including lightweight tactile-token encoding, force-matrix conditioning, gradient accumulation, and regression-tested legacy HDF5 conversion.

TactiWeave-VP *PyTorch · ResNet-18 · Causal TCN*

Visual-Physical Tactile Fusion

- Built a visual-physical temporal-fusion prototype for fabric classification, combining 64-frame grayscale tactile pressure sequences with calibrated force, friction, and contact-area measurements.
- Used a shared ResNet-18 visual encoder and a six-dimensional physical-state MLP with per-frame fusion, causal dilated TCN processing, and attention pooling; the data loader validates aligned sequences and supports chronological splitting.

IsaacSim-Tactile4OpenWorld *Isaac Lab · UIPC · Force Fields*

Tactile Simulation & Force Fields

- Developed a UIPC/libuipc-based compliant tactile membrane in Isaac Sim/Lab and reconstructed tactile-local 3D Fx/Fy/Fz fields from mesh deformation for Piper and Franka manipulation experiments, including force projection, sign calibration, pressure, and shear estimation.
- Built an iterative validation suite covering contact/no-contact, membrane alignment, grasping, lifting, insertion, and lateral shear, while isolating incompatible Isaac Lab 2.1.1/2.3.2 environments and external asset dependencies.

OpenFireAlert *YOLO11 · TensorRT · Jetson*

Wildfire Monitoring & Alerting

- Developed and optimized a YOLO11 fire-and-smoke detector for complex outdoor environments using the D-Fire dataset; established a PyTorch-to-ONNX-to-TensorRT conversion and Jetson deployment pipeline, delivering an edge-vision solution that balances detection robustness, inference efficiency, and deployability.
- Engineered an end-to-end alerting system in which Jetson performs edge inference and publishes annotated RTSP streams and detection states to a Spring Boot/MySQL control plane that fuses ESP32-S3 smoke-sensor signals, suppresses duplicate alerts, and coordinates device-level buzzer, email, and dashboard notifications.

TactileFlowField *PyTorch · RKNN · RV1126B*

Visual-Tactile Dense Optical Flow

- Developed three-stage dual-frame networks for visual-tactile dense optical-flow estimation, exploring feature extraction, cross-frame fusion, and flow-field regression with standardized input, checkpoint, and output interfaces.
- Built a PyTorch streaming-inference pipeline with ONNX/RKNN conversion for RV1126B, supporting FP16/PTQ INT8, per-camera calibration, and multi-device inference under RKNN operator constraints.

EdgeVisTrack-RKNN *RKNN · Cython · RV1126B*

Sparse Multi-Camera Marker Tracking

- Designed a pure-vision multi-camera sparse circular-marker tracking pipeline with regional RGB/Hough initialization, local ROI template correlation, subpixel refinement, and loss recovery, using the CPU implementation as the behavioral reference.
- After profiling five-camera CPU execution at 98%–99% utilization on RV1126B, moved the hot path to Fixed and Dynamic RKNN batch backends and Cython/RKNN C APIs with FP16 LUT conversion, reusable buffers, and stage-level profiling.

Vision Workbench *PySide6 · OpenCV · Release Engineering*

Computer Vision Workbench

- Built a unified PySide6 desktop application exposing seven computer-vision workflows through modular Python APIs, from classical image processing and panorama reconstruction to YOLO detection, segmentation, and training.
- Productized the application with optional dependency isolation, 184 automated test functions, cross-platform Qt/accessibility checks, and verified wheel/Windows EXE updates using version gates, SHA-256 validation, dependency fingerprints, and rollback.

AdaptiveUI-SKILL *Python · Static Analysis · Agent Skills*

Responsive UI Audit & Repair

- Designed two explicit-invocation Agent Skills that turn responsive UI work into an audit-repair-verification workflow across HTML/CSS, React, Vue, Svelte, and Tailwind projects.
- Implemented a zero-dependency Python analyzer with 24 stable rules, severity/confidence metadata, suppressions, JSON Schema output, and CI thresholds, backed by 79 automated tests.

AgentTools *PowerShell · Diagnostics · Codex*

Agent Engineering & Diagnostics

- Built a read-only Codex diagnostics tool covering PowerShell profiles, Conda hooks, Python encoding, Git line endings, proxy/TUN state, and reconnect logs, without exposing credentials or modifying the system.
- Added cross-platform AGENTS.md templates, task handoff, and runtime Skill inventory for explicit-invocation plugins.

Skills

Robot	PyTorch, LeRobot, ACT, PI0/PI0.5, Piper, Visual-Tactile Sensing
Vision	OpenCV, YOLO11, U-Net
Systems	GStreamer, RTSP, MIPI, HDF5, Qt (PySide6/PyQt5), Spring Boot 3, ESP32-S3
Simulation	Isaac Sim, Isaac Lab, UIPC/libuipc
Programming	Python, Cython, Java, C/C++
Edge Deployment	ONNX, TensorRT, RKNN/RKNNLite, FP16, PTQ INT8, Jetson, RV1126B, RK3588